Does AI Debate Produce Better Verdicts?

Benchmark Results — ml-debate-lab

Adversarial debate between AI agents produces measurably more correct evaluations than single-pass assessment — but not for all the reasons we expected.

20 benchmark cases · 3 conditions · known correct answers

The Research Question

Standard AI evaluation is single-pass: give a model some work, ask what it thinks, get an answer.

When does this fail?

- The flaw requires independently questioning the framing
- The work is actually valid but sounds questionable
- The correct answer is "run this specific test first" — not a binary verdict

What we tested: Does adding adversarial structure — a Critic and Defender in debate — produce *measurably more correct* verdicts on cases where we already know the right answer?

> **Key design choice:** Known ground truth for every case. "Better" has a definite answer.

Experimental Design

20 synthetic ML reasoning scenarios across 6 categories:

Category	Description	n
Broken baselines	Flawed comparison methodology	4
Metric mismatches	Wrong metric for the claim	3
Hidden confounders	Temporal, selection, or distribution bias	4
Scope misunderstandings	Conclusion doesn't follow from evidence	4
False-positive traps	Valid work, adversarial framing	5

Three conditions run against every case. Pass criteria set before running anything: mean ≥ 0.65, ≥ 75% cases pass, lift ≥ +0.10 over baseline.

The Three Conditions

Condition	Design	What it isolates
Single-pass baseline	One agent, one call, no structure	The floor — unstructured LLM evaluation
Compute-matched ensemble	3 independent assessors + synthesizer, no roles	What more compute and perspectives alone buy
Debate protocol	Critic → Defender → Judge, sequenced dispatch	What adversarial role structure adds on top

The ensemble uses identical total compute to the debate. Any gap between them is attributable to role structure, not compute budget.

> **Why the ensemble condition?** Without it, you can't separate "debate is better" from "more LLM calls are better."

Results

All three pre-registered benchmark criteria passed.

Condition	Mean Score	Cases Passed
Single-pass baseline	0.38	12 / 20
Compute-matched ensemble	0.75	17 / 20
Debate protocol	**0.97**	**19 / 20**

Raw lift debate vs. baseline: **+0.586**. But two rubric dimensions score structurally differently — the baseline has no Defender (DC = 0.0 by design) and is capped on resolution quality (DRQ ≤ 0.5).

> **Honest corrected lift:** +0.335 to +0.441 after neutralizing structural rubric penalties. Still 3–4× the pre-registered +0.10 threshold. Debate vs. ensemble: +0.216 ($p = 0.004$, $r = 0.758$).

What Debate Does NOT Uniquely Provide

Two claimed advantages turned out to be wrong.

Empirical test design (ETD)

The debate protocol reliably produces well-specified empirical tests. We attributed this to adversarial forcing. An ablation falsified that: adding one explicit output instruction to the ensemble synthesizer achieves ETD mean **0.962** — nearly matching debate's 1.0.

ETD is a prompt design effect. It's portable to any multi-agent configuration.

Exclusive exoneration of valid work

A compute-matched ensemble correctly exonerated valid work in **4 of 5** false-positive trap cases without role isolation. Structural isolation is not uniquely necessary for reaching the right verdict.

> **Implication:** You can get well-specified tests from an ensemble by adding one instruction. The debate's edge is elsewhere.

What Debate DOES Provide

Structured argumentation and contested-position resolution *(confirmed)*

The Critic/Defender structure forces point-by-point rebuttal — every claim conceded, rebutted, or flagged as empirically open. More importantly, it forces engagement with both sides of a genuinely two-sided question.

The clearest example (mm002): An offline NDCG improvement challenged by a concern about offline-online correlation validity.

- Ensemble: all 3 assessors ignored the concern and agreed to run an A/B test → score **0.000**
- Debate: correctly identified that a calibration study was needed first → score **1.0**

This failure mode is **structural to parallel ensemble design**. Adding more assessors doesn't fix it — they all converge on the same intuitive answer because no one is required to argue the other side.

The False-Positive Trap Finding

Five cases: valid work, sound methodology, presented under adversarial framing. The question is whether the protocol wrongly condemns it.

Condition	Correct verdicts	Clean exonerations
Single-pass baseline	**0 / 5**	0 / 5
Compute-matched ensemble	4 / 5	2 / 4 correct
Debate protocol	**5 / 5**	3 / 5

The baseline scored **0.000 on all five** — it accepted the adversarial framing entirely and condemned every piece of valid work it was shown.

The "clean exoneration" edge (3/5 no concerns vs. ensemble hedging on 2) is directional only — n=5 is below statistical threshold and the pattern did not replicate in the external exoneration benchmark.

External Validation

The synthetic benchmark could have been too easy or too tailored. Two separate external benchmarks tested this.

Fault detection — 10 cases from published ML failures
Cases from Dacrema 2019, Obermeyer 2019, DeGrave 2021, and others. Ground truth from the published record — no designer involvement.

→ Debate protocol IDR = **0.95**, meeting the ≥ 0.85 pre-specified threshold.

Exoneration — 3 cases from peer-reviewed ML work
BERT/SQuAD 1.1, ResNet-152/ImageNet, clinical 5-fold CV — methodology is genuinely sound.

→ Debate protocol **3/3 pass** (mean 0.875). Baseline: 0/3 on rubric (DC = 0.0 structural rule), correct verdict label in all 3.

> **Note:** Critics raised plausible-but-wrong concerns on all 3 external exoneration cases (IDP = 0.5). The "clean exoneration" tendency from the internal benchmark did not replicate externally.

⚠️ Remaining Limitations

Take the results at face value, but know what they can't yet tell you.

Limitation	What it means
Synthetic benchmark	20 cases, author-designed. Expanding requires expensive ground-truth labeling.
Same model family	All agents used claude-sonnet-4-6. Cross-vendor validation (GPT-4o, Gemini) is pending.
n=5 exoneration finding	Below conventional statistical thresholds. Directional only.
Author-assigned difficulty labels	Difficulty calibration was not independently verified.
No cross-vendor scorer	Haiku validation confirmed no IDR bias within Anthropic family; external scorer pending.

Bottom line: Results are specific to the claude-sonnet-4-6 capability tier. The honest corrected lift (+0.335 to +0.441) clears the pre-registered threshold by 3–4×, but replicate before treating as general.